

The Gender Gap in STEM — or the Gender Gap in Survival?

Saima Khan | PhD Candidate in Economics | University of Tasmania

The first time I decided to stop parking my salary in a savings account and make some investments, I was positively shocked at the difference it can make. After three hours of going through stock market predictions, which were so elegantly drawn out on graphs, I put my first few dollars in a business I believed would do well. A 10 percent return on average! Not too bad. Thank goodness for some data literacy!

Claiming the world is ‘data-driven’ is an understatement. From financial decisions to making sense of your health, data has taken the front seat, back seat and a few more - and navigating all of it takes math. Math that manifests in everyday decisions and careers. Yet, in spite of this mounting importance and usefulness of math, half the population is raised in a world where they are either told they are not good at it or it’s just not for them. The result? Uninformed decision making for a start, but also [potential talents lost to sciences, technology, finance and economics, and all other fields that are rooted in maths](#) (yes, I mean STEM and beyond), that could quite possibly make this world a better place.

Speaking about the gender gap in math-intensive fields inevitably leads to discussion on fairness and equity, but this argument clearly doesn’t ring a bell as important enough given we have only come so far in tackling this issue. Efficiency is a rather fit argument. One that we need to care about regardless of whether we think the world is fair towards women or not.

Economics 101 would tell us the marginal woman in STEM is more talented than a marginal man – in other words, in a group of ten men solving a problem, a woman would add substantially more value than an additional man – and given the woman obviously had to overcome more barriers, it makes the argument even stronger.

Nevertheless, female representation remains alarmingly limited. This isn’t driven by one uninformed decision, [but multiple decisions taken over time shaped by the environment we are brought up in](#). We often believe we are making rational decisions but these decisions reflect what we absorb from the environment and believe is better for us. Refraining from quantitative electives in university? [Maybe grade 12 mathematics only had ten girls in the classroom out of twenty eight](#). Avoided that mathematics classroom? [Perhaps a teacher once told us math](#)

[wasn't working out for us after we failed a test](#). We probably failed in the first place because we got anxious making sense out of a subject that feels profoundly remote. The majority of us grew up with dolls anyway, while building blocks were kept aside for the boys. From toys that we first play with to the [TV shows we watch](#), to the careers we finally get to, are we truly making free choices or are we responding to years of signals?

Sure, there are policies and interventions researchers rack their brains over. When you come to think of it, though, they tell us “*You can do it too!*”. Here’s a female economist – “*You can become one too!*” Female engineer? “*You can be like her!*” Well, with such bare minimum representation, that’s a hard sell.

An invitation laced with inspiration is not sufficient. Role models are great but not quite enough to uplift the entire population because some of us might have decided not to invest all our energy into becoming things that work against us. What may work, though, and research has shown too, is compelling reasons of survival. Think of STEM not as professions but as a skillset for survival. STEM, which essentially reflects analytical thinking and data literacy, is a fundamental survival skill. And we can most certainly make a case for acquiring skills that are needed for survival. Frame data literacy as a means to daily decision making. [Research shows that when girls are shown the relevance of STEM subjects to their own life goals, their engagement and persistence measurably improve](#).

A tiny nudge to download an investment guide, a push to learn coding - and the ripple effect on women and the economy could be enormous – more diversity, higher efficiency. That’s a recipe for economic growth.

Finally, getting women to be data-literate will naturally open doors for [STEM jobs because what even is a STEM job anymore?](#) A journalist needs to read statistics. A policy analyst needs to model scenarios. A nurse needs to interpret data dashboards. The “STEM job” is increasingly just... any job. If you add daily activities to the list, STEM skills are basically survival skills. It’s time we understood that. As for the policymakers, there’s only so much inspiration can do – but a reason for survival? That’s unavoidable.